

AMR SENTINEL

TEAM X | SHOBHIT UNIVERSITY

AGENDA

Mitigating Outpatient Antimicrobial Resistance (AMR) via Point-of-Care Multi-Agent Clinical Prescription Auditing and Guideline Enforcement

Index

- AMR
- CORE PROBLEM/CAUSE
- ACTION PLAN
- IMPACT
- SUMMARY



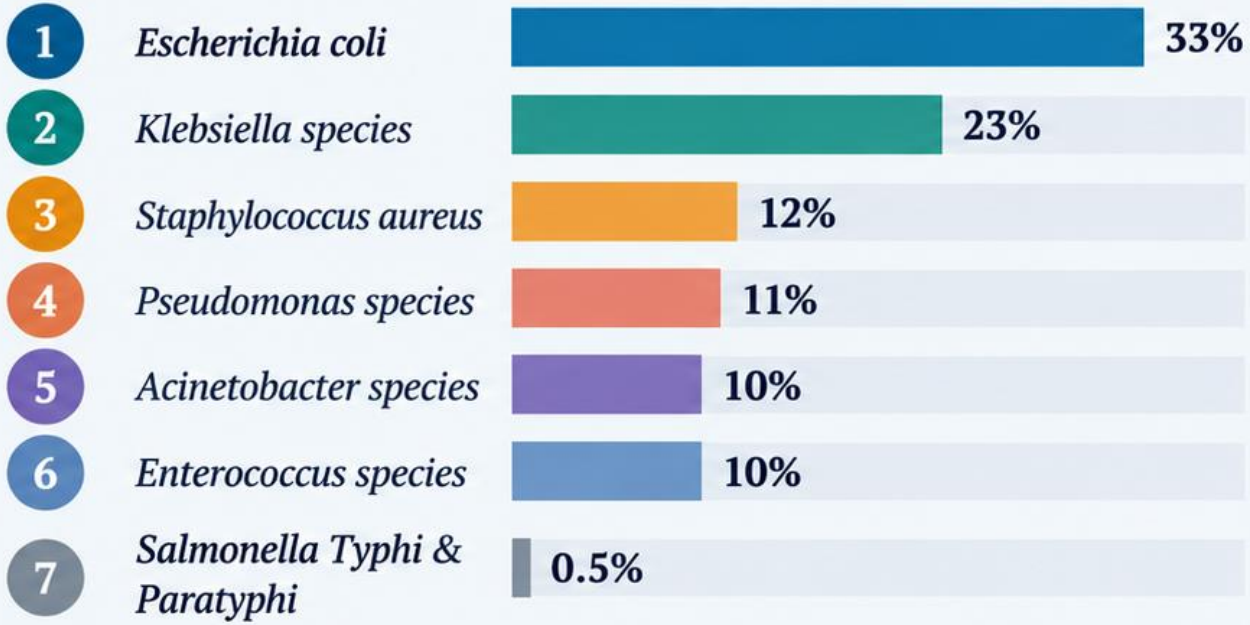
AMR

India AMR Surveillance: Frequently Identified Pathogens & Most-Consumed Antibiotics

NCDC reports | 2024

MOST FREQUENTLY IDENTIFIED PATHOGENS

NCDC NARS-Net 2024 | 195,077 unique-patient isolate records



MOST-CONSUMED ANTIBIOTICS

NCDC Antimicrobial Consumption Surveillance 2024

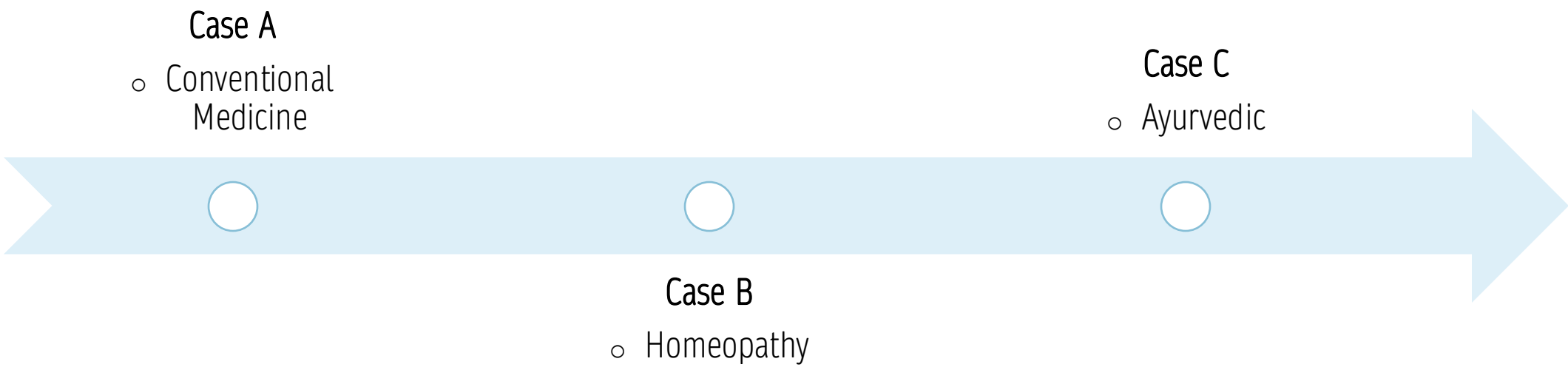
Rank	Antibiotic	DDD per 100 bed-days	WHO AWaRe category
1	Azithromycin	25.1	WATCH
2	Amoxicillin + clavulanic acid	23.1	ACCESS
3	Doxycycline	11.8	ACCESS
4	Ceftriaxone	11.3	WATCH
5	Cefixime	11.0	WATCH

Sources: NCDC, National AMR Surveillance Network (NARS-Net) Annual Report 2024; NCDC, Antimicrobial Consumption Surveillance Report 2024.

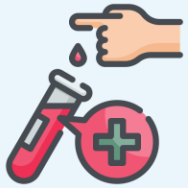
Notes: Pathogen percentages are shares of reported surveillance isolates, not population infection rates.
Consumption is a separate measure; it does not establish which antibiotic caused resistance in a pathogen.

CASE STUDY

This study will talk about three cases for when a patient is diagnosed with UTI



COMPETITIVE LANDSCAPE



Conventional

- Recurrent MDR Klebsiella pneumoniae UTI in a kidney-transplant recipient.
- Treatment failure required reassessment and surgical source control.

Homeopathic

- Two published cases of antibiotic-resistant recurrent UTI.
- Reported improvement; controlled evidence remains insufficient to establish efficacy.



Ayurvedic

- E. coli UTI treated with Varividaryadi Kashaya in a single-patient report.
- Follow-up culture was reported negative; AMR was not established.

CORE PROBLEM

- Point-of-Care Bottleneck
- ICMR Standard-Point Blank
- Overprescription
- OPD: Race Against Time
- Cognition X Manuals

ACTION PLAN



- CLINIC AI AGENT
- PRESCRIPT GUIDANCE
- EFFICIENCY X EFFECTIVITY
- SUSTAINABILITY



TECH STACK (ACTION PLAN 0.1)

FRONTEND

- SHAD CN
- FRAMER MOTION
- NEXT.JS
- TAILWIND CSS
- RECHART
- LUCIDE

BACKEND

- PYTHON
- FASTAPI
- PYDANTIC
- SQLALCEMY
- PSYCOPG
- POSTGRESQL
- PYTEST

AI & RAG

- LANGGRAPH
- LLAMA INDEX
- GEMINI 2.5 FLASH
- OPENROUTER API
- PGVECTOR
- RANK_BM25
- BAAI/BGE-RERANKER
- RAPIDFUZZ
- OKF

IMPACT & FUTURE SCOPE

- **Reducing Irrational Antibiotic Prescribing**
  Prescription audit
- **Slowing the Spread of Antimicrobial Resistance**
  Avoidable Anti-microbial exposure
- **Smarter Healthcare System**
  Practicality and productivity
- **Real Time Surveillance**
  Hospital & Regional Integrations
- **Clinical Personalization**
  Patient Specific Audit
- **AMR Intelligence**
  Awareness and Stewardship

SUMMARY

- AMR
- India's AMR Surveillance
- WHO AMR Guard
- AMR Sentinel
- Impacts & Scope

SOURCES

- WHO
- NCDC
- ICMR
- HRI
- JETIR.ORG
- INTERNATIONAL JOURNAL OF HOMEOPATHIC SCIENCE
- SCIENCE DIRECT
- PUBMED
- AYU (AN INTERNATIONAL QUARTERLY JOURNAL OF RESEARCH IN AYURVEDA
- PIB
- AYUSH



THANK YOU

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